Session 5 — Project Tasks and Submission Checklist – P4A 2026

Complete these tasks in your group notebook before the submission deadline. This is the final project session — all deliverables are due after today.

What you covered today

- LLMs in practice with Ollama and LangChain: prompt templates, multi-turn conversations, structured output
- Local agents: wrapping Python functions as tools, building a reasoning agent, reading the trace

Tasks

1. Interpret your findings with an LLM

Pass a summary of two findings from your Session 4 analysis to a local LLM and ask it to interpret what they mean together. Choose a system prompt and user prompt that fit the business context of your analysis.

Include in your notebook:

- The data you passed to the model
- The prompts you used and why you chose them
- The model's response
- One sentence: does the response add value beyond what the data already shows, or is it just restating the numbers?

2. Structured output

Ask the LLM to summarise one finding from your project as a structured JSON object. Design the fields yourself — they should be useful for filtering, sorting, or comparing results programmatically.

Include in your notebook:

- The prompt you used
- The raw model response
- The parsed output

- One sentence: did the model return valid JSON on the first try? If not, what did you do?

3. Multi-turn conversation

Hold a 3-turn conversation about your project findings. Design the turns yourself — they should build on each other, moving from a broad observation to something specific and actionable.

Include in your notebook:

- All three questions and all three responses
- The message history printed at the end (roles and content previews)
- One sentence: did the model maintain context correctly across the three turns?

4. Build a data agent

Define at least one tool that queries your project data — it must be different from the tools built in the classroom exercise. The tool should return something useful for answering a question about the dataset.

Build an agent. Show the full trace.

5. Finalise `analysis.py`

Your `analysis.py` must contain at least the following functions before submission:

- `load_postings(engine)` — loads the `postings_with_benefits` staging table and returns a DataFrame
- `filter_fulltime(df)` — filters to full-time postings and returns a DataFrame
- `company_summary(engine)` — loads the `companies_companies` table, filters to US companies, and returns a DataFrame
- `top_industries(df_postings, df_job_industries, df_industries, n)` — returns the top N industries by posting count as a DataFrame

6. Review your notebook before submission

Check that:

- All cells run from top to bottom without errors
- Every chart has a title and labelled axes
- Exploratory or draft cells are removed — the notebook tells one clear story
- Markdown cells explain each section in plain language

Submission Checklist

Code (submitted as .zip via Virtual Campus)

- [ ] staging.sql — creates the staging tables
- [ ] analysis.py — all required functions
- [ ] notebook.ipynb — all outputs visible, includes the Text-to-SQL prompt, examples, and discussion

Slides (submitted as PDF via Virtual Campus)

- [ ] 8–15 slides
- [ ] Covers all three project phases: SQL, Python, AI
- [ ] Includes at least one chart from the Python analysis
- [ ] Includes the LLM interaction and your critical assessment of its output
- [ ] Includes the agent task: tool definition, question asked, and trace
- [ ] Includes dataset attribution (dataset slide or footer)
- [ ] Submitted as PDF

Deadlines

Class	Deadline
Class 2_2	26 May 2026, 23:59

Both deliverables must be submitted via Virtual Campus before the deadline.